

Space Propulsion 2024-25

Project assignment

The frontier of cubesat propulsion: MEMs units

Introduction

Minaturization of space propulsion systems is important for full exploitation of nano/pico satellite family, including cubesats and pocket-size sats. When Delta V budgets are below about 2 m/s in total, MEMs propulsion systems represent an innovative solution.

MEMs propulsion is still not consolidated. It is a matter of research and development. Currently, the development revolves around testing of engineered solutions: basically, it is passing through the transition between industrial research and the final application.

MEMs stands for micro-mechanical systems. In propulsion, it indicates those units which very small size. They represent an appealing solution for few-U cubesats, picosats, pocketQubes, and similar. They employ complex and accurate production technology. They are characterized by small amount of propellant, featuring small impulse bits and high mass ratios.

Typical MEMs propulsion subsystems can be based on solid or liquid propellant technology, monopropellant, cold/warm gas thrusters. There is a wide range of solutions, with different degrees of complexity and of performance.

Objective of the activity

The team has to build up a background on MEMs and develop a comparative design among different propulsion solutions. The number of requested solutions depends on the number of team members. The students have to document the following items:

- nominal design
- power to run the unit
- volume requirements of the propulsion unit and of the tanks if applicable
- intrinsic uncertainty correlated to the propulsion system architecture (in particular, the accuracy of most important geometric details)
- comparatively comment the results, motivating your choice for a cubesat mission of 3-U, with a total Delta V of 0,5 m/s and a nominal thrust of 5 mN.

Regarding the Monte Carlo propagation, the model will aim at understanding how nozzle construction accuracy and respective uncertainty can propagate towards the thrust profile (single impulse bit and thrust).

As a general rule for fluids, consider the existence of a volume (plenum) before the injection. This volume has an inlet (from a regulated flow) and an outlet (the injector) and has the task to recover all the pressure. Inside the plenum you have locally a total pressure.

Requirements

Nominal thrust: 5 mN

Total delta V (referred to a 3U cubesat): 0,5 m/s (single or multiple engines, single or multiple firings, at your choice)

Nominal chamber pressure: to be selected from the literature. Nozzle for in-space operations.

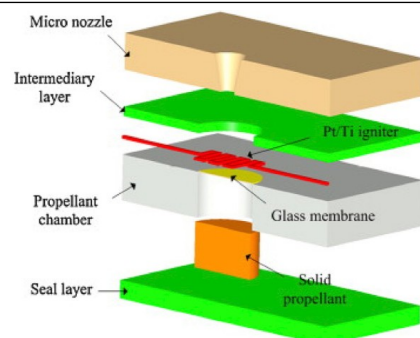
Technologies to explore:

- Teams up to 6 members: 2 configurations (solid + vapor expansion)
- Teams up to 8 members: 3 configurations (solid + vapor expansion + cold gas)
- Teams up to 10 members: 4 configurations (solid + vapor expansion + cold gas + bipropellant)

Tecnologies

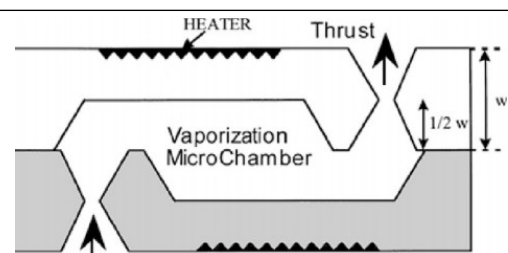
Solid propellant: propellant based on AP/HTPB/Al, as for the BARIA motor (check flipped class assignment for propellant details).

Suggested assumptions: cylindrical grain, constant area burning, no 2-phase flow, adiabatic walls, neglect the igniter, neglect thermal management, isentropic nozzle expansion, calorically perfect gas.



Vapor expansion: water based propulsion system with vaporizer and nozzle, are similar to the figure.

Suggested assumption: neglect friction, consider calorically perfect gas, no power dispersion (power generation of heaters goes into the flow), inlet at standard conditions, remember to consider vaporization enthalpy and sensible enthalpy increment, isentropic flow from the exit convergent, size the nozzle and the upstream plenum, consider the presence of a pressure regulator and of a valve, injection via Cd computation (if not available for specific size, use $C_d = 0.7$), stay below metal operating temperature. Use the input power to the flow as your



variable: do not further detail it with current/voltage relations or dispersion. Check typical values in the literature.

Bipropellant: hydrazine / nitrogen tetroxide hypergolic reaction inside a miniaturized combustion chamber

Suggested assumption: neglect friction, consider calorically perfect gas, consider no heat dispersion, remember to consider vaporization enthalpy and sensible enthalpy increment, isentropic flow from the exit convergent, size the nozzle and the two upstream plenums, consider the presence of two independent pressure regulators and of two valves, injection via Cd computation (if not available for specific size, use $C_d = 0.7$). Stay below metal operating temperature. Flame temperature computed using thermodynamics.



Cold gas thruster: nitrogen-based expansion through a plenum, a filter, and a valve.

Suggested assumption: neglect friction, consider calorically perfect gas, isentropic flow from the exit convergent, size the nozzle and the upstream plenum, consider the presence of a pressure regulator and of a valve, injection via Cd computation (if not available for specific size, use $C_d = 0.7$). Inlet at standard conditions.



MOOG model 058E143

Discussion of the assignment

Here is a suggested path of progress.

A literature survey on MEMS technology should supply to you information about typical sizes, performance, production techniques. It is very important to derive nozzle efficiencies of such engines.

The design should be performed using the techniques learned in class. Some of them are at their limit of validity due to microfluidics: you can neglect this and use them anyway.

Report a schematics of each solution and, on that basis, compute feeding line and size the chamber. Derive information from the literature to support your sizing. Nozzle should be simple: only a convergent-divergent conical nozzle. Remember to add 2D loss.

For the Monte Carlo you should discuss the sensitivity of nozzle throat diameter accuracy on final performance. For each technology, consider these uncertainties:

- Solid: nozzle diameter (from the literature of micro-component manufacturing), Vieille's exponents a and n (from BARIA data or other experimental uncertainty data)
- Vapor expansion: nozzle diameter (from the literature of micro-component manufacturing), liquid injector diameter, power generation
- Bipropellant: nozzle diameter (from the literature of micro-component manufacturing), liquid injector diameters.
- Cold gas thruster: nozzle diameter (from the literature of micro-component manufacturing), liquid injector diameter.

Consider that the plenum before each igniter is ideal so it supplies you a constant pressure, regardless of the mass flow rate. Remember that the chamber mass flow rate will change and, through C^* , the combustion chamber pressure will change. A nested loop should be implemented, as both injectors mass flow rate and nozzle depend on the chamber pressure.

Structure of the assignment

The report will be 25 pages max and will have to include:

- A brief literature analysis
- The modeling approach (essential collection of equations with references, no theory)
- The results
- A comment to the results

Outside the 25 pages, the students can attach an appendix that can complement the text, but it is not essential for the comprehension of the work (tables, sources of data from other books, etc.).

In addition, an authorship declaration is mandatory. The declaration is considered accepted by the whole team. Each member has to be enlisted for at least one technical task. Literature, programming, modeling, etc. are technical tasks. Writing/revising the report is NOT a technical task. One task may be allocated to more than one member.

Randomly, members of groups will be selected and people have to know, at least in general forms, the work done by the team (general modelling approach, main hypotheses and conclusions).

Example of authorship declaration

MEMBER A: contribution to model of tanks, internal ballistics, thermal protection sizing

MEMBER B: literature survey of solid propellants

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Resources

An initial set of references is loaded on WEBEEP. You can start from there.

Other details

Delivery deadline April 9th at 23:59 CEST. Collection will be done through FORMS and will be opened in the next few days.

GET PREPARED in advance for delivery. One representative from each group will be in charge of submission: one team one submission only. This person becomes the contact point of the team.

Name the project file TEAMNAME-PersonalCodeContactPoint.pdf.

During submission, the representative will have to write in the form:

- The name of the team (your choice)
- Full name of each single member
- Personal code (the one 10XXXXXX) of each single member
- Contact email of each single member
- Delivery of project in PDF form (max 10 MB)